Proceedings of the 5th International Conference on Inventive Research in Computing Applications (ICIRCA 2023)
IEEE Xplore Part Number: CFP23N67-ART; ISBN: 979-8-3503-2142-5

Improving the Restaurant Recommendation Performance using Sentiment Analysis

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2023 5th International Conference on Inventive Research in Computing Applications (ICIRCA) | 979-8-3503-2142-5/23/$31.00 ©2023 IEEE | DOI: 10.1109/ICIRCA57980.2023.10220869

***Abstract*— Recommendation systems are effective information filtering systems that filter relevant information from a large amount of dynamically generated data and then provide prioritized information to the customers based on their preferences, likes and other factors. In this study, machine learning based restaurant recommendation system is developed and its performance is improved utilizing the reviews given by the customers. The Zomato Bangalore dataset used in this study is downloaded from Kaggle. Initially, reviews of various restaurants are given as input to classifiers namely Multinomial Naive Bayes (MNB) and K - Nearest Neighbor (KNN) to build models that can classify a given review as positive or negative. Then, similarity measures like Pearson correlation coefficient and cosine similarity are used to identify the reviews that are similar. Finally, the top ranked restaurants are then recommended to the user. Experiments were conducted to evaluate the performance of machine learning algorithms and similarity metrics. From the results obtained, it is found that MNB classifier with cosine similarity is able to recommend restaurants with higher accuracy.**

Keywords — Restaurant Recommendation, Multinomial Naïve Bayes classifier, K-Nearest Neighbor classifier, Pearson Correlation Coefficient, Cosine Similarity.

I. INTRODUCTION

Machine learning is a branch of study that deals the task of making the machines to learn from past data on their own. Machine learning employs a variety of algorithms to perform classification and predict outcomes based on previously acquired data or extensive knowledge. Image identification, audio recognition, email filtering, Facebook auto tagging, and recommended systems are just a few of the applications. When new data is received, a machine learning model learns from the previous data and finds the results. The availability of large amount of data supports the creation of a good model that more correctly identifies the output.

Machines can comprehend both written language and speech and execute various tasks. Chatbots are a well-known application of NLP. They assist support teams in resolving issues by automatically interpreting and responding to typical language queries. In a word, the goal of Natural Language Processing is to make machines understand human language, which is complex, ambiguous, and immensely diverse.

Due to the growth of the web, the process of filtering and identifying relevant information from a large amount of data has become a vital problem to the users. Information retrieval systems provided a partial solution to this information overload problem but failed to present prioritized and personalized results to the user. Recommendation systems are effective information systems that provide the filtered information to the user based on the preferences and the information's relevancy [1, 2].

The emotional tone of a piece of text may be determined using a Natural Language Processing (NLP) approach. This method is useful for businesses to identify the opinions about a product. Online text on websites and social media platforms are now regarded as a rich source of implicit data. The identification of these remarks and classification of the opinions can reveal the user's preferences. Natural language processing technologies is used to process and identify the hidden feelings of the customer present in the comments to achieve this goal. Review analysis is a strong marketing tool that allows product managers to better understand client emotions and incorporate them into their marketing strategies.

The approaches used for building restaurant recommendation systems are: 1. Demographic Filtering - Every user receives broad suggestions based on best-selling goods and/or cuisine. Users with comparable demographic characteristics are recommended with the same eateries by this system. 2. Content Based Filtering – The recommendations are done based on a certain item. For restaurants, this system makes suggestions based on item information such as menu, location, customer service, aesthetics, price, rating, reviews and so on. 3. Collaborative Filtering - This method connects people who share similar interests and makes suggestions based on that connection. Collaborative filters, unlike their content-based equivalents, do not require item information [3-5].

In Content-Based Filtering, suggestions are based on the description and qualities of the items that the user has previously consumed. To put it another way, these

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algorithms attempt to propose things that are comparable to those that a user previously enjoyed (or is examining in the present). Specifically, a large number of candidate items are compared to previously rated items by the user and the best-matching items are recommended. The objective of this study is to develop a content-based recommender system in which when a restaurant name is entered, the recommender system looks at the reviews of other restaurants and recommends additional restaurants with similar reviews, sorting them from the top rated. The contributions of this study are as follows:

i. Apply count vectorizer and TFIDF weighting schemes and compare their performance.
ii. Compare the performance of KNN and MNB in classifying the reviews as positive or negative.
iii. Use cosine similarity and Pearson correlation coefficient to find the similar reviews and compare their performance.

II. Literature Survey

A tourist recommendation system that customizes suggestions based on user's interests is proposed by [6]. To discover tourist's preferences, the reviews are initially preprocessed and then they are clustered semantically. Similarly, the features of these points of interest are extracted from all user's aggregated reviews about an attraction. Finally, the suggested recommendation system contrasts a user's preferences with features to offer the most relevant places of interest to the user. The authors analyzed that Consumers are very often confused about what to eat. Customers require some assistance and motivation [7, 8]. As a result, a variety of apps, services, and programmes have built food-related recommendations. These include not only the most popular use cases of food, meal, recipe, and restaurant recommendations, but also substitute items, menus, and diets. Like in field of healthy lifestyle, where consumers have more specific dietary requirements and objectives, the latter is especially significant.

Analyzing customer reviews one by one, on the other hand, is a challenging operation that becomes nearly impossible when the quantity of reviews is large. Consequently, it becomes feasible to have a mechanism that automatically evaluates customer reviews and delivers the necessary information accurately and efficiently. The authors of [9] have introduced an NLP-based opinion mining methodology for automatically analyzing consumer opinions. To do so, a captive portal is initially deployed to collect client feedback. Then, using the opinion mining technique, the reviews are analyzed to see how customers feel about food quality, service, environment, and so on. To assess the proposed methodology, a data-driven experiment is carried out. The amount of user comments on websites and social networks has increased dramatically With the growth and advancement of these platforms. The comments provide a valuable source of information that can be analyzed to gain insights into people's preferences across various areas, particularly in relation to food. The Internet of Things paradigm can find multiple applications in extracting people's preferences and [10] presents a method specifically focused on determining individuals' food preferences by analyzing their comments. The method involves extracting food names from individual comments, clustering them together, and conducting sentiment analysis for each extracted food name. Experiments have been conducted using comments from the Trip Advisor website. In this case, 100 people were chosen, and their comments were collected from January to September of 2018.The suggested approach was trained using data and then tested using data from the last three months. The findings show that the suggested method extracts users' food preferences with a high degree of precision.

A methodology for creating taxonomies from a corpus of text documents automatically is proposed in [11]. A part-of-speech parser is used to extract terms from documents in this system. Initially, the terms are filtered. The remaining terms indicate taxonomic concepts. To organize these concepts into a hierarchical structure, two methods are employed: the extended subsumption approach, which considers ideal ancestors to determine the parent of a concept, and a hierarchical clustering algorithm that utilizes multiple text-based window and document scopes to identify concept co-occurrences. In the realm of management and economics, it is found that when picking one of these methodologies, factors like taxonomy quality and depth must be considered. For shallow taxonomies, the subsumption approach is preferred, whereas for deep taxonomies, the hierarchical clustering methodology is recommended.

Tourist companies and experts alike are concerned about customer loyalty in tourism. Lately, technology, including CRM and social networks, has emerged as crucial catalysts for enhancing loyalty in the tourism industry. The platform known as POST-VIA 360 provides comprehensive support for the entire lifespan of tourist loyalty, extending beyond the initial visit. The system uses pervasive ways to capture data from the first visit. POST-VIA 360 analyses data and generates accurate after-visit data, these positioning and bio-inspired recommender systems utilize the obtained information to generate pertinent recommendations [12]. The growing participation of end users in generating web content has opened up a new and substantial source of data regarding users' preferences and aversions. In the realm of marketing decision-making, especially in the context of advertisement recommendations, having access to users' input on social media platforms and extracting consumer insights is of utmost importance. By analyzing this data, advertisement recommendation systems can enhance their understanding of users' interests and preferences, enabling them to provide more tailored suggestions.

A content advertisement recommendation system is developed that takes advantage of data generated by harnessing the input provided by users on social networking sites [13]. This method is supported by a shared ontology model that represented both users' profiles and advertisement content. Natural language processing techniques are employed to construct vectors that capture ontological entities from textual content. These vectors serve as representations for both users and advertisements.

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[14-17]. A sentiment-based food classification framework is proposed which consists of two tasks: sentiment classification and four-course meal classification [18]. In the four-course meal classification, the reviews are categorized into four distinct groups: soups and salads, appetizers, main courses, and desserts. Within each course, extract the top n food items that are favored by the majority of consumers. Learning curves are then evaluated, taking into consideration the trade-off between bias and variance. Support vector machines (SVM) classification technique beats other strategies, according to their findings.

III. Proposed Study

The objective of this study is to create a content based recommendation system using sentiment analysis. To do so, the Zomato Bangalore dataset downloaded from Kaggle is used. In addition to features like location, dish_liked, cuisines and so on, the dataset also consists of reviews of the restaurants. Out of 5000 randomly chosen instances, only 4236 samples are considered for further processing. The dataset consists of 2412 unique restaurants. For training 3389 instances (80%) and for testing 847 instances (20%) are used.

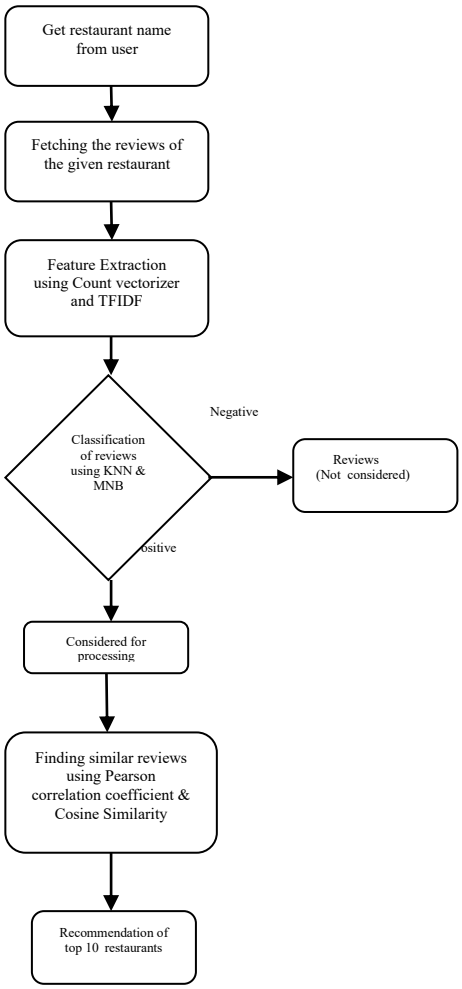

Fig 1. Flow Diagram of the proposed Recommendation System

Initially a restaurant name is received from the user. The reviews of that restaurant is fetched and given as input to classification algorithms like KNN and MNB. The KNN model classifies data by finding the k nearest matches in the training data and then predicting using the labels of the closest matches. The MNB algorithm is a popular Bayesian learning approach. The positive reviews are extracted and similar reviews of are identified using cosine similarity and Pearson Correlation Coefficient. Finally, top 10 restaurants are recommended to the user. The proposed system involves the following steps including:

1. Loading the dataset:
   - Load the data and import the libraries.
2. Data Cleaning:
   - Deleting redundant columns.
   - Renaming the columns.
   - Dropping duplicates.
   - Cleaning individual columns.
   - Remove the NaN values from the dataset
3. Text Preprocessing
   - Cleaning unnecessary words in the reviews
   - Removing links and other unnecessary items
   - Removing Symbols
4. Review Analysis
5. Content based Recommendation
6. Evaluating the performance

Review Analysis

This step begins by analyzing sentiment and categorizing each comment as good or negative. Second, comments are automatically sorted using text classification algorithms based on input regarding meal flavor, ambience, service, and value for money. In this study, with the help of comments and the ratings it splits the positive and negative reviews. After the split, with positive reviews, the recommendation system recommends the restaurant.

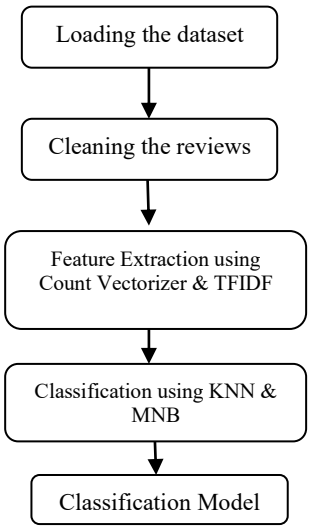

Fig 2. Steps involved in building a classification model

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Count Vectorization

Countvectorizer is a wonderful Python function from the scikit-learn library. It's used to turn a text into a vector depending on the frequency (count) of each word in the text. CountVectorizer tokenizer breaks down a phrase, paragraph, or any text into words and does some basic preprocessing such as removing punctuation marks and turning all words to lowercase.

TF-IDF

The TF-IDF method has an additional inverse frequency term to give importance to the rare words/characters in the document. It assigns a weight to each character based on the frequency with which it appears in all of the documents provided. This lessens the significant impact of too common words on vector output.

IV. Results and Discussion

The machine learning model is being trained using K-Nearest Neighbor algorithm and Multinomial Naive Bayes algorithm. The algorithm that provides the highest accuracy is considered as the best model. From the figure 3, it is found that the K-Nearest Neighbor algorithm yields less accuracy when compared to Multinomial Naive Bayes. Multinomial Naive Bayes shows better performance because for discrete features (for example, word counts in a text classification task). It is the most commonly used model and it operates mostly with integer counts that are generated as frequency for each word.

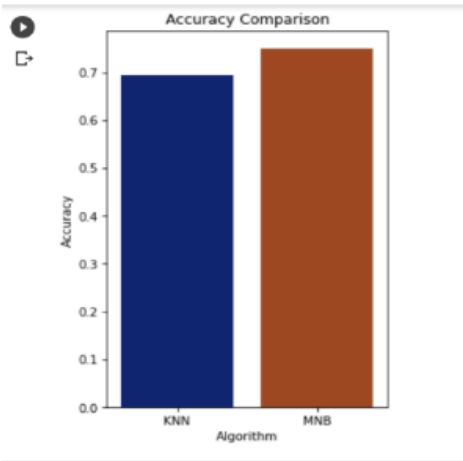

Fig 3. Performance Comparison of the Classifiers

Following the analysis, the reviews are divided into positive and negative reviews, with only the positive reviews proceeding to the next phases, while the negative reviews are ignored because the negative reviews will promote restaurants with the same qualities when cosine similarity is done. And also, if the negative reviews are neglected, time duration for recommending the restaurant will be reduced. TF-IDF is used to extract features from positive evaluations. The recommendation engine then selects restaurants with comparable reviews using cosine similarity, and the top 10 eateries are recommended. Also, the Recommendation System recommends the restaurant nearer to the given input restaurant location as shown in Fig 3. The location of the given input is Jayanagar. After the recommendation as in Fig 4, it suggests 3 restaurants in that location.

Fig 4. Sample Input (Restaurant Name)

	cuisines	Mean Rating	location
Meghana Foods	Biryani, North Indian, Chinese	4.19	Jayanagar
Halli Jonne Biriyani	Biryani, Kebab	4.00	Jayanagar
Meghana Foods Grand	Biryani, Seafood, Andhra, North Indian	4.00	Marathahalli
Rajanna Military Hotel	Biryani, Fast Food	3.86	Jayanagar
Basmati Delights	Biryani, Chinese, Mughlai, Fast Food, Kebab	3.71	BTM
Pramukh Family Restaurant	North Indian, Chinese, Mughlai	3.57	Basavanagudi
The Biryani Paradise	Biryani, South Indian	3.43	Richmond Road
Aasife & Brothers Biriyani Centre	Biryani, North Indian, South Indian, Chinese	3.43	HSR
Green Hyderabad Biryani House	Biryani, Hyderabadi, Andhra, North Indian	3.43	Whitefield
Shanmukha	Biryani, Andhra, North Indian, Chinese	3.36	Old Airport Road

Fig 5. Sample Recommendation of Restaurants

V. Conclusion and Future Work

A content based restaurant recommendation system using sentiment analysis is proposed in this study. From the experiments conducted, it is found that TFIDF weighting scheme performs batter than Count vectorizer. Among the classifiers used for review classification, Multinomial Naïve Bayes classifier performs better than KNN. Also,cosine similarity metric is able to choose the similar reviews better than Pearson Correlation Coefficient metric.

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